can be set up as unit tests with test doubles rather than much slower and harder-to-set-up integration tests.

- Try to avoid high-cost activities like calling external libraries and saving a lot of objects to a database. In testing, a good way to avoid this is to use test doubles to prevent unit tests from having to use real dependencies.
- If you find that a single bug makes multiple tests fail, think about whether all those tests are needed. If the failure is in setup, think about whether all those tests need all that setup.
- Sometimes tests that are useful during development are completely superseded by later tests. These tests can be deleted.
- If it takes a lot of setup to write a unit test, consider the possibility that the test is trying to tell you that the design of the code could be improved to minimize dependencies.

Prescription 11 Think about both the short-term and long-term cost of tests as you write them.

SWIFT: The Five Qualities of Valuable Tests

A valuable test saves time and effort over the long term, while a poor test costs time and effort. I've focused on five qualities that tend to make a test save time and effort. The absence of these qualities is often a sign that the test could be a problem in the future. I've even managed to turn them into an acronym that is only slightly contrived: SWIFT:

- Straightforward
- Well defined
- Independent
- Fast
- Truthful

Let's explore those in more detail.

Straightforward

A test is *straightforward* if its purpose is immediately understandable.

Straightforwardness in testing goes beyond just having clear code. A straightforward test is also clear about how it fits into the larger test suite. It should test something different from the other tests, and that purpose should be easy to discern from reading the test. That purpose is usually tied